

Chan Junheng

Phonetics & Phonology, Computational & Algorithmic Thinking

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Education

M.A. (by research) in Linguistics & Multilingual Studies

2022 - 2024

Nanyang Technological University (NTU), Singapore

Advisor: Assoc. Prof. Scott Reid Moisik

Thesis: *Quantitative analysis of voice quality: The influence of the protruded jaw setting on labiodental fricatives*

- Conducted principal component analysis (PCA) of magnetic resonance imaging (MRI) data from 48 participants, demonstrating that 31% of productions shifted from labiodental to dentolabial articulation under a protruded jaw setting, contributing empirical evidence for the principle of segmental susceptibility in voice quality theory.

B.Sc. (Hons) in Mathematical Sciences (Highest Distinction)

2018 - 2022

Nanyang Technological University (NTU), Singapore

Specialisation in Applied Mathematics

Supervisors: Prof. Wang Li-Lian and Assoc. Prof. Scott Reid Moisik

Final Year Project: *A Study on Vocal-Ventricular Fold Coupling with Glottal Airflow Simulation*

- Developed a mathematical framework for incorporating realistic airflow into vocal tract simulation in ArtiSynth using the lattice Boltzmann method (LBM), investigating the conditions for and potential effects on self-sustained vocal fold oscillation and associated oscillatory patterns in vocal-ventricular fold coupling (VVFC).

Research Experience

Research Associate, LanTERN Project (NTU)

2025 - Present

- Conduct phonetic analysis of Siyuewu Khroskyabs, examining phonological structure, acoustic patterns, and documentation data for ongoing descriptive and experimental linguistic research.
- Investigate language contact in Singaporean Hokkien and Malay through corpus-based documentation, phonetic analysis, and comparative study of contact-induced variation.

Participant, NTU-Undana Language Description and Linguistic Analysis Workshop

2025

- Conducted elicitation-based research on Hawu language with native speaker consultants, collecting and analysing primary linguistic data for language documentation.
- Received training in linguistic field methods, transcription, and documentation using FieldWorks, ELAN, and SayMore, applying these tools to the collaborative analysis of Hawu data.

Research Assistant, Transcription of Singaporean Hokkien (NTU)

2021 - 2022

- Transcribed a Singaporean Hokkien speech corpus into IPA and Written Hokkien (閩南漢字 Bân-lâm hàn-jī) using ELAN annotation software.
- Produced time-aligned annotations supporting corpus development and future research on phonetics, lexicon, and documentation of an under-described Southern Min variety.

Research Intern, A*STAR I²R

2021

Project: Glottal Features for Potential Improvement of Child-Adult Voice Conversion

- Investigated acoustic features capturing nonlinear source-filter interactions in child speech, motivated by child-adult anatomical differences affecting phonation and speech production.
- Performed Linear Discriminant Analysis using TensorFlow, improving child-adult class separation and increasing classification performance in decision tree modelling.

Research Intern, DSO National Laboratories

2020

Project: Natural vs Spoofed Speech Classification using Voice Source Features

- Investigated voice source features complementing MFCC and CQCC representations for spoofed speech detection using Voice Analysis Toolkit and COVAREP repository for feature extraction.
- Developed a spoofed speech detection algorithm by training extracted features in a Gaussian Mixture Model classifier from the ASVspoof 2019 Challenge.

Undergraduate Researcher, URECA (NTU)

2019 - 2020

Project: Characterising Patterns of Glottalisation in Singaporean English

- Conducted auditory and acoustic analyses of Singapore English glottalisation using the National Speech Corpus and Praat waveforms and spectrograms.

- Developed an LSTM-based detection algorithm for glottal stops and creaky voice using MATLAB Deep Learning Toolbox and segmented speech data.

Conference Presentations

2026

- Chan, Junheng. "The Influence of the Protruded Jaw Setting on Labiodental Fricative /v/." Poster presentation at LabPhon 20, Montreal, 2026. [Forthcoming]
- Chan, Junheng and Mustapha Kamal, Nur Amirah Hakim. "Consonant Correspondences of Malay Loanwords in Singaporean Hokkien." Poster presentation at the workshop *When Chinese Meets Southeast Asian Languages: Two Millennia of Continuous Language Contact*, SEALS 35, 2026.
- Chan, Junheng. "The Influence of Coda Stops on Tones in Singaporean Hokkien." Oral presentation at SEALS 35, 2026.
- Chan, Junheng and Mustapha Kamal, Nur Amirah Hakim. "Stress-Tone Correspondences of Malay Loanwords in Singaporean Hokkien." Oral presentation at SEALS 35, 2026.

Teaching & Mentoring Experience

Teaching Assistant, NTU Department of Linguistics & Multilingual Studies (LMS)

2023 - 2024

- Courses:
 - HG2001 Morphology and Syntax (2024)
 - HG2020 Language in Society (2024)
 - HG1001 Mind and Meaning (2023)
 - HG2003 Phonetics and Phonology (2023)
- Planned and taught up to 12 weekly tutorial classes for approximately 200 students across linguistics modules.
- Developed supplementary learning materials with teaching teams and assisted in quiz design and assessment.

Peer Mentor, NTU SPMS Learning Centre

2020 - 2021

- Provided academic mentoring to 20 students across 14 mathematics modules, supporting coursework and conceptual understanding outside scheduled classes.
- Guided students in navigating undergraduate study in mathematics through individual consultation and peer mentoring.

Peer Mentor, NTU WSC Deaf Community

2018 - 2020

- Provided weekly academic mentoring for two Deaf and hard-of-hearing students preparing for N-Level examinations.
- Organised outreach events including SPARKZ 2019 and Christmas Song Sign to strengthen mentorship and promote awareness of Singapore's Deaf community.

Awards

College of Humanities, Arts, and Social Sciences Teaching Assistantship

2022 - 2024

School of Humanities Teaching Award (SoTA)

2023

Tan Sri Dr Tan Chin Tuan Scholarship

2021 - 2022

Tan Sri Dr Runme Shaw Scholarship

2019 - 2020

Skills

Software: Python, R, MATLAB, Praat, ELAN

Languages: English, Mandarin, Hokkien